

MHC-I Peptide-MHC Binding Prediction Using SVM and BLOSUM62 Encoding

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Repository: <https://github.com/varadha0505-max/mhc-binding-predictor>

Abstract

Developed a supervised machine learning model to predict peptide binding to the MHC Class I molecule HLA-A02:01, a key step in neoantigen-based personalised cancer vaccine design. Using 8,346 experimentally validated peptide-MHC binding measurements from the Immune Epitope Database (IEDB), we trained a Support Vector Machine (SVM) with an RBF kernel on BLOSUM62-encoded 9-mer peptide features. The model achieved ROC-AUC 0.949 and 88.6% accuracy on a held-out test set, exceeding our target of $AUC > 0.85$ and approaching the performance of published tools such as NetMHCpan. Permutation-based feature importance analysis revealed that position P2 carries the highest predictive weight, and that Leucine at P2 is the single most important amino acid feature — consistent with the known HLA-A02:01 anchor residue preference (P2=Leu/Met) established by crystallographic and biochemical studies. This biological finding emerged from sequence data alone, without any prior structural supervision.

1. Introduction

1.1 Biological Background

Cancer immunotherapy exploits the ability of CD8⁺ cytotoxic T-lymphocytes (CTLs) to recognise and kill tumour cells. Recognition depends on a three-component interaction: a peptide fragment derived from an intracellular protein, presented by an MHC Class I molecule on the cell surface, detected by a T-cell receptor. Tumour-specific somatic mutations can generate novel peptide sequences (neoantigens) not present in the normal proteome. If these neoantigens bind MHC-I with sufficient affinity ($IC_{50} < 500$ nM by convention), they can trigger a tumour-specific immune response.

Personalised cancer vaccines (currently in Phase III trials by Moderna/BioNTech for melanoma) work by identifying patient-specific neoantigens from tumour sequencing data and synthesising mRNA encoding those peptides. The bottleneck is computationally predicting which of thousands of candidate peptides will actually bind MHC — wet-lab validation of every candidate is infeasible.

1.2 HLA-A*02:01

HLA-A*02:01 is the most common MHC Class I allele in humans (~45% population frequency). It has the most curated binding data in IEDB and is the primary allele studied in neoantigen vaccine trials. It has known anchor preferences: hydrophobic residues (Leu, Met) at position 2, and hydrophobic residues (Val, Leu) at position 9 of 9-mer peptides.

1.3 Problem Statement

Given a 9-amino acid peptide sequence, predict whether it binds HLA-A*02:01 with $IC_{50} < 500$ nM (binary classification).

2. Data

2.1 Source

Immune Epitope Database (IEDB), queried for:

- MHC Restriction: HLA-A*02:01
- Assay type: IC_{50} (nM)
- Organism: Homo sapiens
- Downloaded: July 2026

2.2 Cleaning Pipeline

Step	Action	Rows remaining
Raw download	—	30,146
Filter nM IC_{50} assays	Remove non- IC_{50} assay types	17,096
Drop missing values	Remove NaN peptide or IC_{50}	17,075
Keep 9-mers	Filter by peptide length == 9	11,723
Standard amino acids	Remove non-canonical residues	11,723
Deduplicate	Median IC_{50} per unique peptide	8,346

2.3 Label Definition

- Binder: $IC_{50} < 500$ nM (positive class, label=1) — 3,112 peptides (37.3%)
- Non-binder: $IC_{50} \geq 500$ nM (negative class, label=0) — 5,234 peptides (62.7%)

The 500 nM threshold is the standard used in NetMHCpan and the majority of MHC binding prediction literature.

3. Methods

3.1 Feature Engineering

One-Hot Encoding (OHE):

Each peptide is represented as a 9×20 binary matrix where position $(i,j)=1$ if amino acid j occupies position i . Flattened to a 180-dimensional vector. Treats all amino acids as equally dissimilar.

BLOSUM62 Encoding:

Each amino acid is replaced by its row in the BLOSUM62 substitution matrix — a 20-dimensional vector of log-odds substitution scores derived from evolutionarily conserved protein blocks. This encodes physicochemical similarity: amino acids with similar biochemical properties receive similar vectors. Each 9-mer

becomes a 9×20 matrix (180-dimensional vector after flattening), normalised to zero mean and unit variance before model training.

BLOSUM62 was chosen over OHE because MHC binding pockets select residues partly by size and hydrophobicity — properties that BLOSUM62 implicitly encodes through evolutionary conservation patterns.

3.2 Model

Support Vector Machine (SVM) with Radial Basis Function (RBF) kernel, implemented via scikit-learn's CalibratedClassifierCV wrapper for probability estimates.

Hyperparameters selected by 5-fold cross-validated grid search over:

- $C \in \{0.1, 1.0, 10.0, 100.0\}$
- $\gamma \in \{\text{scale}, 0.001, 0.01\}$

Optimal: $C=10$, $\gamma=0.001$.

Class imbalance handled by `class_weight='balanced'`.

3.3 Evaluation

- Primary metric: ROC-AUC (threshold-independent, appropriate for imbalanced data)
- Secondary: Accuracy, Precision, Recall, F1, Average Precision
- Train/test split: 80/20 stratified
- Cross-validation: 5-fold stratified

3.4 Interpretability

Permutation importance: each of the 180 features was randomly shuffled 10 times; the mean decrease in ROC-AUC was recorded. Features were reshaped to a 9×20 matrix (position $\times$ amino acid) for biological interpretation.

4. Results

4.1 Model Performance

Metric	Value
ROC-AUC	0.9486
Average Precision	0.8968
Accuracy	88.6%
Non-binder Precision	0.92
Non-binder Recall	0.90
Binder Precision	0.83
Binder Recall	0.87

Metric	Value
CV AUC (5-fold)	0.934 ± 0.009

4.2 Biological Validation — Anchor Residues

Permutation importance analysis revealed:

- P2 is the single most important peptide position
- Within P2, Leucine (L) has the highest importance score of any amino acid at any position in the entire 9×20 feature matrix
- P9 shows elevated importance relative to central positions

This precisely matches the experimentally determined anchor residue preferences of HLA-A*02:01: P2=Leucine/Methionine (primary anchor), P9=Valine/Leucine (secondary anchor). The model recovered this structural constraint from IC50 measurements alone.

4.3 Neoantigen Panel

Peptide	Source	Score	Result
GILGFVFTL	Influenza M1	0.943	✓
NLVPMVATV	CMV pp65	0.929	✓
RMFPNAPYL	HPV / Cervical CA	0.881	✓
KILDGVQKL	Melanoma BRAF V600E	0.840	✓
YLQLQHFNL	Melanoma NRAS Q61R	0.840	✓
FLTPKKLQC	Negative control	0.041	✓
DDDPGIARI	Negative control	0.002	✓
GGGPGGGGG	Negative control	0.000	✓
KLGGALQAK	Influenza M1 (weak)	0.053	X*

*KLGGALQAK is a documented weak binder with IC50 ~2,000 nM, above the 500 nM classification threshold. The model's low score is consistent with this borderline status.

Panel accuracy: 8/9 (89%)

5. Discussion

5.1 Performance

AUC 0.949 substantially exceeds the project target of 0.85 and is competitive with published tools. NetMHCpan 4.1 reports AUC ~0.96 on benchmark datasets, trained on significantly more data and using neural network architectures. The gap is consistent with our simpler model and smaller training set.

5.2 Why BLOSUM62 Works

The MHC binding groove selects peptides based on the physicochemical complementarity between residue side chains and groove pockets. BLOSUM62 encodes this implicitly: amino acids that substitute for each other in evolution tend to be biochemically similar. OHE treats all substitutions as equally wrong; BLOSUM62 treats similar amino acids as similar features, which is biologically appropriate.

5.3 Limitations

1. Allele-specific: model only applies to HLA-A*02:01
2. Binary threshold (500 nM) loses affinity information; regression on log-IC50 would be more informative
3. No structural features (peptide flexibility, MHC groove geometry)
4. KLGGALQAK miss highlights boundary cases near the threshold

5.4 Future Directions

1. Pan-allele model: concatenate HLA pseudo-sequence with peptide encoding
2. Regression target: predict log-IC50 directly
3. Deep learning: 1D-CNN or Transformer encoder for positional motif learning
4. Application: screen TCGA somatic mutations to rank candidate neoantigens
5. Deployment: Streamlit web app for clinical/research use

6. Conclusion

Built a complete ML pipeline for MHC-I peptide binding prediction achieving AUC 0.949 on 8,346 HLA-A*02:01 peptides. The model's feature importance profile independently recovered the known P2 Leucine anchor preference — a structural constraint established by decades of biochemical and crystallographic research — demonstrating that BLOSUM62-encoded SVM models can extract meaningful biological signal from binding affinity data. This validates the approach as a foundation for more sophisticated neoantigen prioritisation pipelines.

References

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